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ABSTRACT

Large language models (LLMs) produce fluent enterprise answers but remain prone to hallucination, exposing organisations to operational, legal and reputational risk. Retrieval-Augmented Generation (RAG) grounds generation in retrieved evidence, yet its reliability is bounded by the retriever, and many systems apply a single retrieval paradigm, or a fixed fusion weight, to every query. This dissertation asks whether adaptive hybrid retrieval and citation-grounded generation make enterprise question answering measurably more reliable. A single reproducible Python framework evaluates eight configurations: sparse (BM25), dense (all-MiniLM-L6-v2, FAISS), a static Reciprocal Rank Fusion hybrid, a fixed-weight hybrid and a query-adaptive hybrid, the three hybrids each with and without inline citations, across FinanceBench, CUAD and a custom Azure corpus. Generation used GPT-4.1 at temperature zero; answers were judged with RAGAS (GPT-4o), and configurations compared per question using Wilcoxon signed-rank tests, bootstrap confidence intervals, Cohen's d_z and Holm–Bonferroni correction.

The findings are more nuanced than the guiding hypothesis anticipated. Hybrid retrieval significantly outperformed sparse retrieval on FinanceBench (Recall@5 rising from 0.30 to 0.59; $p < 0.001$) but did not surpass a strong dense retriever (Recall@5 0.68). Adaptive weighting significantly improved retrieval ranking over a fixed-weight hybrid (MRR and nDCG@5 up by approximately 0.067; $p \approx 0.0004$), but this gain did not reach answer-quality metrics or the other corpora. Citation grounding did not significantly improve faithfulness on any comparison (a small, direction-varying effect), although it produced a zero invalid-citation rate and moderate-to-high coverage.

The central finding is that retrieval-ranking gains from adaptive fusion do not automatically become answer-faithfulness gains; inline citation is best treated as an auditability feature rather than a hallucination remedy. A confound-controlled comparison, a transparent adaptive-weighting mechanism and citation-coverage measures enable this dissociation to be observed cleanly.

Keywords: Retrieval-Augmented Generation, Hybrid Retrieval, Adaptive Query Routing, Citation Grounding, Faithfulness, Enterprise Question Answering, RAGAS, Information Retrieval.

TABLE OF CONTENTS

ABSTRACT 2

LIST OF ABBREVIATIONS 5

LIST OF TABLES 5

LIST OF FIGURES 5

1. INTRODUCTION 6

1.1 Background 6

1.2 Problem Statement 6

1.3 Research Objectives 7

1.4 Research Contribution 7

2. LITERATURE REVIEW 9

2.1 From Closed-Book Generation to Retrieval-Augmented Generation 9

2.2 Sparse, Dense and the Case for Hybrid Retrieval 9

2.3 Adaptive and Query-Dependent Retrieval 10

2.4 Answer Verifiability and Citation Grounding 10

2.5 Evaluating RAG End to End 11

2.6 The Research Gap 11

3. METHODOLOGY 13

3.1 Research Design and Approach 13

3.2 System Architecture 13

3.2.1 Chunking and Indexing 14

3.2.2 Retrieval and Fusion 14

3.3 The Adaptive Router 14

3.4 Generation and Prompting 16

3.5 Datasets 16

3.6 Evaluation Metrics 17

3.7 Statistical Analysis 19

3.8 Reproducibility and Ethics 19

4. RESULTS AND ANALYSIS 21

4.1 Aggregate Performance 21

4.2 RQ1: Does Hybrid Retrieval Beat Single-Method Retrieval? 22

4.3 RQ2: Does Citation Grounding Improve Faithfulness? 23

4.4 RQ3: Does Adaptive Weighting Beat a Fixed Weight? 25

4.5 Router Behaviour 25
4.6 Error Analysis 26
5. DISCUSSION 28
5.1 Interpreting the Findings Against the Hypothesis 28
5.2 Why Retrieval Gains Do Not Become Faithfulness Gains 28
5.3 The Meaning of the Citation Result 28
5.4 The Router's Corpus Dependence 28
5.5 Methodological Considerations 29
5.6 Threats to Validity 29
5.7 Comparison with the Literature 29
6. CONCLUSION 31
6.1 Summary of Contributions 31
6.2 Implications for Practice 31
6.3 Limitations 31
6.4 Future Work 32
6.5 Concluding Remarks 32
REFERENCES 33
APPENDIX A: CONFIGURATION AND HYPERPARAMETERS 36
APPENDIX B: DATASETS 36
APPENDIX C: DIGITAL RESOURCES AND CODE REPOSITORY 36

LIST OF ABBREVIATIONS

BM25	Best Matching 25 (sparse lexical retrieval function)
CUAD	Contract Understanding Atticus Dataset
FAISS	Facebook AI Similarity Search
GPT	Generative Pre-trained Transformer
IR	Information Retrieval
LLM	Large Language Model
MAP	Mean Average Precision
MRR	Mean Reciprocal Rank
nDCG	normalised Discounted Cumulative Gain
QA	Question Answering
RAG	Retrieval-Augmented Generation
RAGAS	Retrieval-Augmented Generation Assessment
RRF	Reciprocal Rank Fusion
SEC	United States Securities and Exchange Commission

LIST OF TABLES

Table 2.1 Positioning of this study against comparable retrieval, adaptive, citation and evaluation work 12
Table 3.1 The eight evaluated configurations and their role in the RQ1–RQ3 design..... 13
Table 3.2 The seven query features computed by the adaptive router 16
Table 3.3 The three evaluation corpora: scale, source and role 17
Table 3.4 Evaluation metrics, formulae and research-question mapping..... 19
Table 4.1 FinanceBench aggregate results (means)..... 21
Table 4.1b The three evaluation regimes 21
Table 4.2 RQ1, FinanceBench: static hybrid (RRF) versus sparse BM25 22
Table 4.3 RQ2: effect of citation grounding on faithfulness 24
Table 4.4 RQ3, FinanceBench: adaptive weighting versus fixed-weight hybrid 25
Table 6.1 Summary of research-question verdicts 31

LIST OF FIGURES

Figure 3.1 End-to-end architecture of the adaptive hybrid RAG pipeline 14
Figure 4.1 Mean Recall@5 by retrieval strategy across the three corpora 23
Figure 4.2 Mean RAGAS faithfulness by configuration across the three corpora 24
Figure 4.3 Distribution of adaptive-router route labels per corpus..... 26

1. INTRODUCTION

1.1 Background

Large language models are becoming a common interface for reasoning over organisational knowledge: employees increasingly expect to ask a natural-language question (about a financial disclosure, a contractual obligation or a technical configuration) and receive a fluent, immediately usable answer. That fluency is the source of the risk. Contemporary LLMs generate coherent, persuasive text irrespective of whether the underlying claim is true, and remain prone to hallucination: the confident assertion of factually incorrect or unsupported information (Ji et al., 2023). In consumer settings a hallucinated answer is an inconvenience; in the enterprise domains that motivate this study (financial analysis, legal contract review and technical knowledge management), an authoritative but unsupported claim can carry material operational, legal and reputational consequences.

The dominant architectural response is Retrieval-Augmented Generation (RAG), which conditions the model on passages retrieved from an external knowledge base at inference time instead of on parametric memory alone (Lewis et al., 2020). Grounding generation in retrieved evidence reduces factual error relative to closed-book generation (Shuster et al., 2021), and a recent survey positions RAG as the prevailing pattern for grounding LLM applications (Gao et al., 2023). A RAG system's reliability is nonetheless bounded above by what it retrieves: if the evidence before the model does not contain the answer, no amount of careful generation recovers it. Retrieval quality is therefore the central, and frequently under-examined, determinant of end-to-end reliability.

Two design decisions dominate retrieval quality. The first is the retrieval paradigm. Sparse lexical methods such as BM25 excel at exact-term and clause-level matching but are brittle to paraphrase (Robertson and Zaragoza, 2009); dense bi-encoder methods capture semantic similarity (Karpukhin et al., 2020) but can underperform on precise terminology in specialised corpora (Thakur et al., 2021). Because these strengths are complementary, hybrid retrieval is an appealing compromise. The second is how the two signals are fused, and whether that fusion is fixed or query-dependent. Many hybrid systems apply one configuration to every query, a poor fit for enterprise QA, where one interaction may demand exact keyword matching ("What was 3M's FY2018 capital expenditure?") and the next broad semantic interpretation ("why did operating margins decline?"). The enterprise setting sharpens this: knowledge bases are heterogeneous and tolerance for unsupported claims is low. The three corpora used here (financial filings, commercial contracts and technical documentation) span that heterogeneity while remaining publicly reproducible.

1.2 Problem Statement

Prior evaluations of hybrid and adaptive retrieval report retrieval-focused metrics (Recall@K, Precision@K, mean reciprocal rank and nDCG@K) but rarely measure end-to-end faithfulness alongside them (Thakur et al., 2021; Gao et al., 2023). It therefore remains unclear whether improvements in retrieval ranking translate into a more faithful answer. This matters because retrieval and generation fail independently: a retriever may surface the correct passage while the generator synthesises an unsupported claim, or fail while the generator produces a confident but ungrounded answer. Evaluating retrieval in isolation cannot distinguish these cases.

A second, distinct gap concerns verifiability. Citation-supported generation can make outputs auditable and has been linked to improved factual accuracy (Nakano et al., 2021), yet many RAG systems return

answers with no inline attribution, leaving claims hard to verify in regulated contexts where provenance carries audit and legal weight. However, a citation does not guarantee correctness (a system may cite an irrelevant passage or leave a claim unsupported), so citation grounding must be treated as a measurable property, not a presentational flourish. Little published work jointly evaluates adaptive routing and citation-grounded generation in a controlled, enterprise-focused comparison. This dissertation addresses that gap directly. The primary research question is:

To what extent does an adaptive hybrid RAG framework (combining BM25 sparse retrieval, dense vector retrieval and citation-grounded generation) reduce hallucination-related errors and improve answer faithfulness compared with single-method and static-hybrid RAG baselines in enterprise question answering?

1.3 Research Objectives

Three sub-questions operationalise the primary question and structure the empirical work:

● RQ1: Does hybrid retrieval achieve higher retrieval quality than sparse-only or dense-only retrieval across enterprise-style corpora?

● RQ2: Does citation-grounded generation improve answer faithfulness and verifiability relative to non-citation RAG generation?

● RQ3: Does adaptive query routing (adjusting sparse and dense weighting by query type and retrieval confidence) improve retrieval and answer quality over a static hybrid configuration?

● The guiding hypothesis, from the proposal, was that adaptive hybrid retrieval with citation grounding would be more faithful and correct than single-method baselines, and that citations would improve verifiability without reducing relevancy.

These questions map onto concrete engineering objectives: (i) to implement a modular, reproducible pipeline supporting all five retrieval strategies under identical chunking, embedding and generation settings; (ii) to construct a transparent, untrained adaptive router whose every decision is reconstructable from logged features; (iii) to instrument the pipeline with both classical IR metrics and a separate answer-quality judgement; and (iv) to analyse the results with a paired, multiple-comparison-corrected protocol that guards against over-interpretation.

1.4 Research Contribution

The primary contribution is an empirical finding enabled by two methodological choices, each paired with a limitation:

Primary finding: Retrieval gains do not become faithfulness gains. By reporting retrieval metrics (Recall@5, MRR, nDCG@5) alongside RAGAS faithfulness and citation measures within the same paired statistical framework, the study surfaces the retrieval-gain-without-faithfulness-gain dissociation that evaluations measuring retrieval alone cannot observe. This is the study's most substantive empirical finding. Limitation: faithfulness is measured by a separate LLM judge (GPT-4o), but because it shares a provider with the generator (GPT-4.1) this reduces rather than eliminates self-preference bias, and no human annotation was undertaken.

Contribution 1: Transparent adaptive router. The study contributes a fully auditable, untrained adaptive-

weighting mechanism in which every routing decision is a deterministic function of seven logged query features. Unlike learned adaptive methods (Jeong et al., 2024; Asai et al., 2024) that require labelled training data and sacrifice interpretability, this router needs no training set, every decision is reconstructable from the audit log, and it can be deployed by practitioners without a supervised routing dataset. **Limitation:** the heuristic weights are manually specified and shown to help on only one corpus, where queries are heterogeneous; the router's benefit is contingent on query diversity and may not generalise without light tuning.

Contribution 2: Confound-controlled RQ3 design with two static hybrids. The experimental design isolates the adaptivity variable by giving the RQ3 baseline and treatment the identical weighted-fusion function, differing only in whether the dense weight is fixed (0.5) or set per query. This eliminates the fusion-method confound that would arise from comparing RRF against weighted fusion, and allows the effect of adaptivity to be read directly from the results. **Limitation:** only one fixed weight (0.5) was tested as the RQ3 baseline; the adaptive router might not outperform a carefully tuned fixed weight selected on a validation split, so the gain is demonstrated against an equal-weight default, not an optimised competitor.

2. LITERATURE REVIEW

This chapter moves from the general reliability problem to the specific gap this study addresses (Section 2.6).

2.1 From Closed-Book Generation to Retrieval-Augmented Generation

The reliability problem addressed here follows directly from how LLMs store knowledge. Parametric knowledge (facts implicitly encoded in model weights during pre-training) is static, opaque and impossible to update without retraining, and the model cannot signal when a required fact is absent. Ji et al. (2023), surveying hallucination in natural language generation, attribute it to both data-related and modelling-related causes, distinguishing intrinsic hallucination (contradiction of the source) from extrinsic hallucination (unverifiable additions). Their taxonomy motivates this study's central construct: faithfulness, the degree to which an answer's claims are grounded in the evidence supplied to the model.

Retrieval-Augmented Generation was introduced by Lewis et al. (2020) to couple a parametric generator with a non-parametric memory (a retrievable external corpus), so generation is conditioned on evidence fetched at inference time. This decouples knowledge from weights: the corpus can be updated without retraining, and retrieved passages give an inspectable basis for the answer. Shuster et al. (2021) showed empirically that retrieval augmentation reduces hallucination in knowledge-grounded dialogue, establishing the grounding-reduces-hallucination relationship that underpins enterprise interest in RAG. The survey of Gao et al. (2023) identifies retrieval quality as a recurring bottleneck: naive pipelines inherit the weaknesses of whichever retriever they employ, and much advanced-RAG engineering targets retrieval, not generation.

A more recent survey (Fan et al., 2024) confirms this trajectory, classifying advances along the retrieval, generation and augmentation dimensions, while the question of whether a generated answer is genuinely grounded in the passages it cites remains unresolved (Wallat et al., 2025).

2.2 Sparse, Dense and the Case for Hybrid Retrieval

Sparse lexical retrieval, canonically BM25 (Robertson and Zaragoza, 2009), scores documents by weighted exact-term overlap with the query. It is fast, interpretable, needs no training, and is exceptionally strong when the information need uses the same terminology as the corpus, common in legal and financial text, where clause numbers, defined terms and figures must match precisely. Its weakness is equally established: operating on surface form, it fails when the query paraphrases the corpus, retrieving nothing for a semantically correct but lexically divergent question.

Dense retrieval embeds query and passages into a shared vector space where semantic similarity corresponds to geometric proximity. Karpukhin et al. (2020) showed that a bi-encoder dense retriever can substantially outperform BM25 on open-domain QA, capturing paraphrase and conceptual overlap that lexical matching misses. However, the BEIR benchmark (Thakur et al., 2021) showed that dense retrievers do not dominate universally: in zero-shot settings, and particularly on specialised or entity-heavy corpora, BM25 remains competitive or superior, because dense models trained on general text can miss precise terminology. Sparse and dense retrieval thus fail in different, complementary ways, the motivating premise for hybrid retrieval.

Learned sparse retrieval has since matured as a middle ground: Formal et al. (2022) show that distillation-

trained sparse models can match or exceed dense retrievers on in-domain benchmarks while preserving the interpretability of lexical matching, reinforcing the view that no single paradigm dominates. Separately, Cuconasu et al. (2024) show that, in a constrained setting (greedy 15-token answers from 4-bit models), injecting irrelevant 'noise' documents can improve answer accuracy, suggesting that retrieval precision alone is an insufficient predictor of answer quality.

Hybrid retrieval fuses the two ranked lists. Reciprocal Rank Fusion (Cormack, Clarke and Büttcher, 2009) is the most widely adopted method: it sums reciprocal-rank contributions, is parameter-light, and is robust across heterogeneous benchmarks precisely because it discards raw, incomparable scores in favour of ranks. RRF is the canonical hybrid and the RQ1 treatment here. Yet Bruch, Gai and Ingber (2024) show that the choice of fusion materially affects performance, and that score-based convex combinations (weighted fusion) can outperform rank-based fusion when scores are normalised. This is directly relevant: a hybrid's behaviour is governed not only by which retrievers it combines but by how, motivating a weighted fusion function whose single free parameter, the dense weight, can be held fixed or made query-adaptive.

2.3 Adaptive and Query-Dependent Retrieval

The observation that no single retrieval configuration is optimal for every query leads naturally to adaptivity. Self-RAG (Asai et al., 2024) advances adaptive retrieval by training a model to decide, on demand and per segment, whether to retrieve at all and to critique its own generations with reflection tokens. But it targets the retrieve-or-not decision and needs supervised training with special tokens; it does not address query-time weighting between two strategies, the problem framed here. Cross-encoder reranking (Nogueira and Cho, 2019) improves precision by jointly encoding query and candidate for fine-grained relevance scoring, but it is a re-scoring stage applied after retrieval rather than a mechanism for selecting between retrieval strategies, and it adds substantial inference cost. Graph-based approaches such as GraphRAG (Edge et al., 2024) restructure the corpus into an entity graph for global, query-focused summarisation, again a different problem from strategy selection.

Jeong et al. (2024) pursue a complementary direction in Adaptive-RAG, training a lightweight classifier to route queries to no-retrieval, single-step or multi-step retrieval strategies based on question complexity. Their approach demonstrates that adaptive routing can improve accuracy across complexity tiers; however, it requires labelled training data and learns a discrete routing decision rather than a continuous fusion weight. The transparent, untrained router in this study occupies a complementary design point: it trades away learned routing accuracy for full auditability and zero training cost, making it more practical where labelled routing data are unavailable.

Controlled evidence on whether adjusting the sparse–dense fusion weight per query, from transparent features, improves retrieval and end-to-end answer quality in enterprise corpora therefore remains limited, because adaptivity has mostly been learned rather than heuristic. Whether such a transparent router earns its place against a fixed-weight hybrid is a concrete, answerable question the wider literature leaves open.

2.4 Answer Verifiability and Citation Grounding

Verifiability is a second axis of reliability, orthogonal to retrieval quality. Nakano et al. (2021), in WebGPT, showed that requiring a model to collect and cite references while answering can improve factual accuracy

and, equally importantly, renders answers auditable: a human can follow each citation to its source and check the claim. In regulated enterprise contexts this auditability is not cosmetic: provenance carries audit and legal weight, and an answer that cannot be traced to a source may be unusable even if correct.

However, the literature also cautions against treating citation as proof of grounding (Wallat et al., 2025). A model may cite a passage that does not support its sentence, cite a valid passage for an invalid claim or leave load-bearing claims uncited. These are measurable failure modes, and the treatment adopted here is to quantify two of them: citation coverage (the fraction of answer sentences carrying at least one in-range citation) and invalid-citation rate (the fraction of markers pointing outside the supplied passage set). Framing citation as measurement rather than presentation lets the study ask a question usually assumed away: does requiring citations actually change the faithfulness of an answer, or only its presentation?

Wallat et al. (2025) sharpen this point: a model may post-rationalise citations, attaching a plausible source to a claim generated from parametric memory, so citation correctness (the cited passage supports the claim) does not guarantee citation faithfulness (the claim was actually produced from that passage).

2.5 Evaluating RAG End to End

A recurring methodological weakness in the retrieval literature, visible in the evaluation practices surveyed by Gao et al. (2023), is the tendency to evaluate retriever and generator separately, or to judge the generator with reference-overlap metrics that reward surface similarity rather than grounding. Retrieval metrics say nothing about whether the generator used the retrieved evidence faithfully. The RAGAS framework (Es et al., 2024) addresses this by decomposing an answer into atomic claims and using an LLM judge to verify each against the retrieved context, yielding a faithfulness score that operationalises the intrinsic-hallucination construct of Ji et al. (2023), alongside answer relevance; the RAGAS library adds reference-based context recall and answer correctness (Ragas, no date a; no date b). Two cautions attend its use: an LLM judge is itself fallible, exhibiting documented position, verbosity and self-enhancement biases (Zheng et al., 2023), and aggregate faithfulness can conceal locally catastrophic errors. Both are partly mitigated here: the judge differs from the generator, and a stratified error analysis accompanies the aggregate scores.

Beyond RAGAS, alternative evaluation frameworks have been proposed. ARES (Saad-Falcon et al., 2024) automates RAG evaluation with a prediction-powered inference framework that estimates context relevance, answer faithfulness and answer relevance with statistical confidence, reducing the annotation budget. Tamber et al. (2025) benchmark LLM faithfulness in RAG settings with evolving leaderboards, indicating that faithfulness remains a fragile quality dimension. These developments support the multiple evaluation angles adopted here.

2.6 The Research Gap

Three strands of the literature converge on a specific, unfilled gap. First, hybrid retrieval is well studied, but the effect of the fusion function (and of making it adaptive) on end-to-end faithfulness, as opposed to retrieval ranking alone, is rarely measured. Second, adaptive retrieval has been pursued mainly through learned mechanisms that trade away interpretability and require labelled data, leaving the transparent-heuristic design space under-examined. Third, citation grounding is widely deployed but seldom evaluated as a measurable property whose effect on faithfulness is tested rather than assumed. Within the literature

reviewed for this study (surveyed across arXiv, the ACL Anthology, the ACM Digital Library and NeurIPS proceedings via Google Scholar, combining terms for retrieval-augmented generation, hybrid and adaptive retrieval, citation grounding and faithfulness), no work was identified that jointly evaluates a transparent adaptive router and citation-grounded generation in a controlled, enterprise-focused comparison reporting retrieval and faithfulness together. This dissertation occupies exactly that gap, with a design (described next) that isolates each variable so the effects of adaptivity and of citation can each be read directly.

Table 2.1 positions this study against the most directly comparable work: none of these combines a transparent, untrained adaptive router with measured citation grounding and joint retrieval–faithfulness reporting.

Study	Retrieval	Adaptive?	Inline citations?	End-to-end faithfulness?	Key limitation
Karpukhin et al. (2020)	Dense	No	No	No	General-domain; weak on exact terms
Thakur et al. (2021)	Sparse & dense	No	No	No	Zero-shot IR only; no generation
Cormack, Clarke and Büttcher (2009)	Hybrid (RRF)	No	No	No	Rank fusion; retrieval only
Asai et al. (2024)	Dense + critique	Retrieve-or-not	Yes (ASQA)	Partial	Requires training; no sparse–dense weighting
Jeong et al. (2024)	Routed strategies	Learned router	No	Accuracy	Labelled data; opaque routing
Nakano et al. (2021)	Web retrieval	No	Yes	Accuracy (trained)	RLHF-trained; not a fusion study
Es et al. (2024)	– (evaluation)	–	–	Yes (metric)	Provides a metric, not a retriever
This study	Sparse+dense hybrid	Heuristic (untrained)	Yes (measured)	Yes (RAGAS + citation)	Single dense model; LLM-judge proxy

Table 2.1. Positioning of this study against comparable retrieval, adaptive, citation and evaluation work. '–' denotes not applicable.

3. METHODOLOGY

This chapter specifies the controlled experiment and justifies each design choice.

3.1 Research Design and Approach

This study adopts a quantitative, experimental design. A controlled comparative approach suits research questions about performance differences attributable to specific architectural components, which only a design holding all other variables constant can support. Eight configurations were evaluated on identical questions within each corpus, so that any observed difference reflects the retrieval strategy or citation setting rather than incidental variation in chunking, embedding, generation or evaluation.

The eight configurations form a structured two-factor comparison across retrieval strategy (sparse, dense, static hybrid via RRF, fixed-weight hybrid, adaptive hybrid) and citation grounding, with citation and no-citation crossed over the hybrid configurations. The proposal specified six; during implementation this was refined to eight by adding a fixed-weight hybrid, with and without citations. The naive test of adaptivity, adaptive weighted hybrid versus RRF static hybrid, confounds two changes: rank- versus score-based fusion, and fixed versus adaptive weighting. A fixed-weight hybrid using the identical weighted-fusion function as the adaptive treatment, with the dense weight held at 0.5, makes the only variable distinguishing RQ3 baseline from treatment whether that weight is fixed or query-adaptive. Table 3.1 sets out the eight configurations and the research question each serves.

Configuration	Retriever	Fusion	Cites	Role
sparse_bm25	BM25	—	No	RQ1 baseline
dense	Dense (FAISS)	—	No	RQ1 baseline
static_hybrid	BM25 + Dense	RRF (k=60)	No	RQ1 treatment / RQ2 base
static_hybrid_cite	BM25 + Dense	RRF	Yes	RQ2 treatment
static_hybrid_weighted	BM25 + Dense	Weighted (0.5)	No	RQ3 baseline
static_hybrid_weighted_cite	BM25 + Dense	Weighted (0.5)	Yes	RQ3 baseline (cited)
adaptive_hybrid	BM25 + Dense	Weighted (per-query)	No	RQ3 treatment / RQ2 base
adaptive_hybrid_cite	BM25 + Dense	Weighted (per-query)	Yes	RQ3 treatment (cited) / RQ2 treatment

Table 3.1. The eight evaluated configurations and their role in the RQ1–RQ3 design.

3.2 System Architecture

The framework is a single, modular Python pipeline in which a corpus flows through chunking, indexing, retrieval, fusion, generation and evaluation. Clearly separated components let each stage be varied independently while every configuration reuses the same underlying indexes. Figure 3.1 depicts the end-to-end flow.

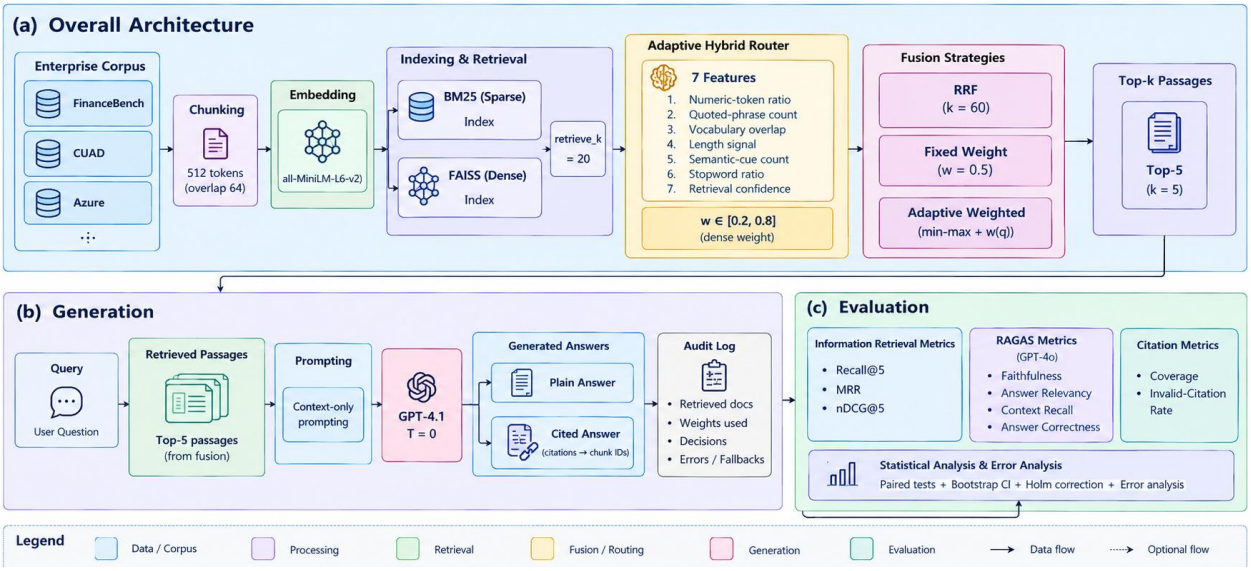


Figure 3.1. End-to-end architecture of the adaptive hybrid RAG pipeline: a corpus is chunked and indexed once (BM25 and dense FAISS), the two ranked lists are fused (RRF, fixed-weight or per-query adaptive router), and the top-k passages are passed to GPT-4.1 for plain or cited generation before IR, RAGAS and citation evaluation. Source: author's own work.

3.2.1 Chunking and Indexing

Documents are segmented into overlapping token windows using a tiktoken cl100k_base tokeniser (chunk size 512 tokens, overlap 64), falling back to whitespace tokenisation where the tokeniser is unavailable. These values are held constant across configurations so none benefits from a different granularity of evidence. Two indexes are built once per corpus and reused by every configuration: a BM25 index (rank-bm25 Okapi) over tokenised chunks, and a dense FAISS index over L2-normalised sentence-transformer embeddings, so inner-product search equals cosine similarity.

3.2.2 Retrieval and Fusion

Sparse retrieval scores chunks by BM25; dense retrieval embeds the query with all-MiniLM-L6-v2 and searches the FAISS index. The static hybrid fuses the two ranked lists with Reciprocal Rank Fusion (k = 60), a widely used rank-fusion method. RRF is deliberately score-agnostic: it combines each item's reciprocal ranks across the lists, so a chunk ranked highly by either retriever rises without the two retrievers' incomparable raw scores ever being placed on one scale. This robustness makes RRF the canonical hybrid and the appropriate RQ1 treatment.

The weighted hybrid takes a score-based route necessary to express adaptivity. Because BM25 scores and cosine similarities occupy different ranges, each retriever's scores are min–max normalised into a comparable [0, 1] range; a chunk's fused score is then a convex combination, (1 − w) times its normalised sparse score plus w times its normalised dense score, governed by a single dense_weight w. At w = 0 the hybrid is pure sparse and at w = 1 pure dense, so w is one interpretable dial from lexical to semantic. The fixed-weight hybrid sets w = 0.5; the adaptive hybrid sets w per query via the router of Section 3.3. All configurations retrieve a candidate pool (retrieve_k = 20) before fusion and return the top k = 5 passages, so fusion always operates over a pool wider than the final set.

3.3 The Adaptive Router

The study's principal design artefact is a transparent, untrained router that sets the dense weight per

query. A heuristic router was chosen deliberately: it needs no labelled routing dataset, and every decision can be reconstructed exactly from logged features, which supports auditing and error analysis. For each query the router computes seven features signalling whether the information need is lexical or semantic (a refinement of the proposal's six: its combined 'numerical and section tokens' signal became a single numeric-token ratio, and a vocabulary-overlap feature was added as an explicit corpus-grounding signal):

● Numeric-token ratio: the proportion of query tokens containing digits, boosted when the query contains a structural term (section, clause, article, appendix, figure, table or step); a lexical signal, since figures and clause references demand exact matching.

● Quoted-phrase count: the number of double-quoted spans, signalling explicit exact-phrase intent; a lexical signal.

● Vocabulary overlap: the fraction of query tokens present in the corpus vocabulary; high overlap favours lexical matching.

● Length signal: query length normalised and saturating at twenty tokens; short queries lean lexical, long queries lean semantic.

● Semantic-cue count: occurrences of interpretive cue words (why, how, explain, compare, difference, impact and similar); a semantic signal.

● Stopword ratio: the proportion of stopwords, a proxy for fluent prose and therefore a semantic signal.

● Retrieval confidence: the difference between the dense and sparse normalised top-1-versus-top-2 margins, each $(s1 - s2)/(|s1| + |s2|)$, indicating which retriever is more decisive for this query; this is a relative margin, not a calibrated probability.

#	Feature	Definition	Signal (weight)
1	numeric_token_ratio	Fraction of query tokens containing a digit; +0.2 if a structural term (e.g. 'section', 'clause', 'table') appears	Lexical (0.9)
2	n_quoted_phrases	Count of double-quoted spans in the query (capped at 1)	Lexical (0.6)
3	vocab_overlap	Fraction of query tokens present in the corpus vocabulary	Lexical (0.5); complement → Semantic (0.3)
4	length_signal	min(n_tokens / 20, 1) — saturates at 20 tokens	Semantic (0.5); complement → Lexical (0.3)
5	n_semantic_cues	Count of cue terms (why / how / explain-type), capped at 2, halved	Semantic (0.7)
6	stopword_ratio	Fraction of query tokens that are stopwords	Semantic (0.4)

#	Feature	Definition	Signal (weight)
7	retrieval_confidence	Dense normalised top-1-vs-top-2 margin minus the sparse one; positive = dense more decisive	Semantic (0.3) if positive, else Lexical (0.3)

Table 3.2. The seven query features computed by the adaptive router, with their definitions and fixed contribution weights to the lexical and semantic scores. Every feature value is logged per query, so each routing decision is fully reconstructable.

These features combine, with the fixed weights of Table 3.2, into a lexical and a semantic score. These weights were set a priori from information-retrieval principles rather than tuned on the evaluation data, deliberately avoiding any overfitting of the router to the benchmarks and preserving its interpretability. The semantic fraction (semantic score over the sum of both) is mapped linearly onto the dense weight within a bounded interval [0.2, 0.8], so a strongly semantic query pushes the weight toward 0.8 and a strongly lexical one toward 0.2. For reporting, the query is labelled semantic when its semantic score exceeds its lexical score by more than 0.15, lexical when more than 0.15 lower, and mixed otherwise. Every feature value, the derived scores, the label and the final weight are logged per query.

These features are heuristic indicators rather than validated predictors of retrieval relevance: a long query can still contain exact terminology, while a numeric expression can require semantic interpretation. Their inclusion therefore reflects transparent design hypotheses rather than learned evidence of optimal routing, and the logged feature values (Appendix C) allow future work to assess which signals carry genuine predictive weight.

The audit log shows how these signals interact, and where they mislead. The FinanceBench question on 3M's FY2018 capital expenditure looks like an exact look-up, but its full 29-token wording has a numeric-token ratio of only 0.07, so length and stopword signals dominated: it was labelled semantic (w = 0.58), and the adaptive hybrid missed the gold passage that RRF retrieved. Conversely, the short interpretive question "Is Boeing's business subject to cyclicality?" received the corpus's lowest weight (w = 0.40).

3.4 Generation and Prompting

Generation uses GPT-4.1 (OpenAI) at temperature zero across every configuration, holding the generator fixed so differences are attributable to retrieval and citation settings, not model choice. Decoding is therefore deterministic up to provider-side non-determinism, as the API seed parameter was not pinned; every generated answer is preserved verbatim in the audit log. Two system prompts are used. The plain prompt instructs the model to answer using only the provided context, concisely and factually, and (critically for measuring abstention) to reply with a fixed refusal string when the context lacks the answer. The cited prompt additionally requires every sentence to end with bracketed citation markers identifying the supporting passage(s) and forbids citing passages that do not support the claim. Retrieved passages are formatted as a numbered context block with source labels, so citation markers map back to chunk identifiers.

Temperature zero was deliberate: because every test is paired, sampling noise would inflate the variance of per-question differences and reduce power to detect the small answer-quality effects discussed in Section 5.5 (Appendix A compares decoding settings in related studies).

3.5 Datasets

The data are system-generated outputs evaluated against benchmark question–answer pairs; no human

participants are involved. Two published benchmarks form the core evaluation, with a smaller custom corpus as a robustness check. All corpora use a common JSON-lines format (documents plus questions with reference answers and gold document IDs), so the pipeline is dataset-agnostic. Table 3.3 summarises the three corpora.

FinanceBench (Islam et al., 2023) contributes its 150 publicly released answerable question–answer pairs over SEC filings, a domain where numerical precision and evidence grounding are paramount. The corpus is built from the dataset's evidence passages, with identical evidence text de-duplicated via content hashing so the retriever is not penalised for returning a textually identical "twin". Because evidence strings rather than full filings are used, this approximates an open-book setting; this is reported as a limitation in Section 6.3.

CUAD (Hendrycks et al., 2021) contributes 150 questions drawn with a fixed seed (42) from the expert-annotated test split; the corpus comprises the 102 test-split contracts in full text. Questions are the dataset's clause-extraction prompts and reference answers the extracted clause spans, where lexical precision is critical. The count was fixed at 150, from 1,244 answerable questions, to keep eight configurations under RAGAS judging within budget; the full corpus is retained, so retrieval difficulty is undiminished. One property shapes every CUAD result: the prompts (for example, a request to highlight the parts of "this contract" related to "Document Name") name no contract, because in the original task each question is posed against a single known contract. Used for corpus-wide retrieval, they give the retriever no signal by which to select the correct document, so retrieval is near-impossible by construction, not by weakness in the methods tested. CUAD is therefore retained for its answer-quality behaviour, but its retrieval metrics read as a stress test rather than a fair comparison, and are therefore not treated as evidence for the comparative superiority of any retrieval strategy (Sections 4.6 and 6.3).

Azure is a custom corpus of 55 technical-documentation files (covering Azure services, RAG architecture and enterprise-AI operations) with 50 manually constructed question–answer pairs spanning lexical and semantic types, each linked to its source document. It serves strictly as a robustness check on documentation-style QA, not as equal in evidential weight to the two published benchmarks; it tests whether the same patterns recur in a realistic enterprise knowledge base.

Corpus	Domain	Docs	Questions	Source	Role in the study
FinanceBench	Financial filings (SEC)	144	150	Islam et al. (2023)	Primary benchmark; discriminates between methods
CUAD	Commercial legal contracts	102	150	Hendrycks et al. (2021)	Stress test; retrieval limited by task construction
Azure	Technical documentation	55	50	Author-compiled	Robustness check; saturated (ceiling effect)

Table 3.3. The three evaluation corpora: scale, source and role. A small sample corpus (6 documents, 6 questions) also ships for a no-cost reproducibility check but contributes no results.

3.6 Evaluation Metrics

Retrieval quality is measured, where gold relevance labels exist, with Recall@5, Precision@5, mean reciprocal rank, nDCG@5 and MAP@5 (Thakur et al., 2021). Relevance is assessed at document level: a chunk counts as relevant if its parent document is in the question's gold set. Recall@5, Precision@5 and MAP@5 count each gold document once, but the nDCG@5 implementation credits every retrieved chunk

of a gold document and can therefore exceed 1 (Section 4.4 reports a de-duplicated check). A NaN-not-zero convention excludes questions lacking gold labels from a metric's mean rather than contributing a spurious zero. The cutoff of five is deliberate: it is the number of passages passed to the generator, so retrieval is scored over exactly the evidence the model sees.

Two consequences of this document-level, single-gold labelling determine which retrieval metrics carry independent weight in Chapter 4. Almost every question has exactly one gold document (mean gold documents per question 1.0 on FinanceBench and CUAD, 1.06 on Azure). Under single-gold relevance, two of the five computed metrics are not independent but exact rescalings of others: Precision@5 equals Recall@5 divided by five, and MAP@5 equals MRR (average precision for one relevant item is the reciprocal of its rank). These identities held without exception on FinanceBench and CUAD, breaking only on the few two-gold Azure questions. To avoid presenting three findings as five, Precision@5 and MAP@5 are reported once in the appendix, and the main-text analysis rests on the three non-redundant metrics: Recall@5, MRR and nDCG@5.

End-to-end answer quality is measured with RAGAS (Es et al., 2024) across four dimensions: faithfulness (the proportion of answer claims grounded in the retrieved context, and the primary hallucination indicator), answer relevancy (alignment between answer and question), context recall (coverage of ground-truth content by the context) and answer correctness (agreement with the reference answer). The judge is GPT-4o at temperature zero, distinct from the GPT-4.1 generator; because both are from the same provider and family (OpenAI), this reduces self-preference bias without eliminating it, so the automated scores are triangulated with the error analysis of Section 4.6. For the cited configurations, citation coverage and invalid-citation rate are additionally reported to quantify verifiability for RQ2. Table 3.4 summarises all metrics used.

Refusals were scored like any other answer, and inconsistently: of 868 refusals, 490 received faithfulness 1.0 and 344 received 0.0. Mean faithfulness therefore partly reflects refusal scoring; BM25's FinanceBench mean of 0.535 combines 0.727 on answered questions with 0.467 on refusals.

Metric	Purpose	Formula (source)	Strengths / Limitations	RQ
Recall@5	Coverage: how many relevant documents appear in the top 5	|relevant ∩ top-5| / |relevant| (Manning, Raghavan and Schütze, 2008)	+ Intuitive coverage measure; − ignores rank order within the window	RQ1, RQ3
Precision@5	Purity: fraction of top-5 results that are relevant	|relevant ∩ top-5| / 5 (Manning, Raghavan and Schütze, 2008)	+ Result cleanliness; − under single-gold labelling equals Recall@5 / 5 (redundant)	RQ1
MRR	Rank of the first relevant hit	mean(1 / rank_first_relevant) (Manning, Raghavan and Schütze, 2008)	+ Ideal for single-answer QA; − ignores everything after the first hit	RQ1, RQ3
nDCG@5	Rank-sensitive relevance gain	DCG = $\Sigma\, rel_i / \log_2(i+1)$, normalised by ideal DCG (Järvelin and Kekäläinen, 2002)	+ Rewards higher-ranked relevant items; − designed for graded relevance (this study uses binary)	RQ1, RQ3
MAP@5	Blended precision and recall across ranks	mean(average precision) (Manning, Raghavan and Schütze, 2008)	+ Combines precision with recall; − under single-gold labelling equals MRR (redundant)	RQ1
Faithfulness	Grounding: fraction of answer claims supported by retrieved context	supported claims / total claims (Es et al., 2024)	+ Primary hallucination signal; − LLM-judge-dependent, inherits judge biases	RQ2, RQ3
Answer	On-topic alignment	Cosine similarity of	+ Measures topicality; − does not test	RQ2, RQ3

Metric	Purpose	Formula (source)	Strengths / Limitations	RQ
relevancy	between answer and question	question with questions regenerated from the answer (Es et al., 2024)	factual correctness	
Context recall	Retrieval sufficiency for the reference answer	Reference claims attributable to context / reference claims (Ragas, no date a)	+ Measures whether the retriever surfaced enough evidence; − requires a reference answer	RQ1, RQ3
Answer correctness	End-to-end quality against the reference	Factual overlap + semantic similarity with reference (Ragas, no date b)	+ Holistic quality check; − conflates factuality with phrasing similarity	RQ1–RQ3
Citation coverage	Attribution density across answer sentences	Sentences with ≥1 valid [n] / total sentences (this study)	+ Measures annotation completeness; − does not verify whether the citation semantically supports the claim	RQ2
Invalid-citation rate	Detection of fabricated passage references	Out-of-range markers / total markers (this study)	+ Catches hallucinated citations; − checks only index range, not semantic support	RQ2

Table 3.4. Evaluation metrics used in this study, with formulae, critical assessment and research-question mapping.

3.7 Statistical Analysis

Because every configuration answers the same questions, all comparisons are paired per question, controlling for question difficulty and substantially increasing power. For each pair and metric, the Wilcoxon signed-rank test is primary, chosen because paired metric differences are frequently non-normal and a rank-based test is therefore more robust, with paired t-test and Shapiro–Wilk normality results reported in the statistics file (Appendix C). Effect size is Cohen's dz (the standardised mean of paired differences), and a 95% confidence interval for the mean difference comes from bootstrap resampling (10,000 resamples, fixed seed). Because many comparisons are made, Holm–Bonferroni correction is applied within each dataset to control the family-wise error rate, and significance is judged on the Holm-adjusted p-value, reducing the risk of over-interpreting differences arising from multiple testing. Each dataset forms a single correction family of 32 pre-specified comparisons spanning RQ1–RQ3, including Precision@5 and MAP@5; though reported only descriptively (Section 3.6), these are retained as a deliberately conservative choice. The bootstrap interval and the significance test answer different questions: the interval is not multiplicity-adjusted, so it can exclude zero for a difference the Holm-adjusted p-value treats as non-significant.

Automated metrics are triangulated with a stratified error analysis of low-faithfulness answers, categorised as retrieval failure, routing-associated retrieval failure, unsupported synthesis or citation mismatch. The analysis does not claim automated metrics measure truth perfectly; they are structured indicators of grounding quality. This matters because a high aggregate score can mask serious local errors: in a legal or financial answer, a single unsupported claim can render an otherwise fluent response unusable.

3.8 Reproducibility and Ethics

Reproducibility is a first-class requirement: bootstrap and CUAD sampling use fixed seeds; dependency versions are pinned; and every retrieval decision, prompt, answer, parsed citation and routing decision is written to a per-question audit log, so any result can be reconstructed. The study presents minimal ethical risk: no human participants, no personal data and no sensitive unpublished material, with all corpora public or openly accessible secondary sources. API keys and environment files are excluded from

submission. Principal technical risks (library deprecation, dataset-access changes and LLM rate limits) are mitigated by pinning dependencies, snapshotting datasets and piloting before full runs.

4. RESULTS AND ANALYSIS

This chapter reports all eight configurations over the three corpora. Each configuration answered 350 questions (150 FinanceBench, 150 CUAD, 50 Azure), producing 2,800 generations, every one generated by GPT-4.1 at temperature zero and judged on four RAGAS dimensions by the separate GPT-4o judge. The two published benchmarks carry the evidential weight, with Azure a robustness check.

4.1 Aggregate Performance

Table 4.1 summarises the headline metrics per configuration on FinanceBench, the corpus that discriminates most clearly between methods. Two patterns are immediately visible. First, sparse-only BM25 is markedly the weakest, with Recall@5 of 0.30 and faithfulness of 0.535, well below every configuration with a dense component. Second, the dense retriever is individually strong, with the highest Recall@5 (0.68) and faithfulness (0.84) of any configuration, a result that materially shapes the interpretation of RQ1.

Configuration	Recall@5	nDCG@5	MRR	Faith.	Ans.Corr.
sparse_bm25	0.300	0.247	0.210	0.535	0.177
dense	**0.680**	**0.570**	0.475	**0.837**	**0.344**
static_hybrid	0.593	0.463	0.389	0.793	0.316
static_hybrid_cite	0.593	0.463	0.389	0.767	0.297
static_hybrid_weighted	0.600	0.493	0.413	0.756	0.299
static_hybrid_weighted_cite	0.600	0.493	0.413	0.800	0.304
adaptive_hybrid	0.633	0.560	**0.480**	0.814	0.310
adaptive_hybrid_cite	0.633	0.560	**0.480**	0.806	0.319

Table 4.1. FinanceBench aggregate results (means; RAGAS faithfulness and answer correctness from the GPT-4o judge). Best value in each column shown in bold.

On CUAD the absolute metrics are uniformly low (Recall@5 between 0.05 and 0.09, answer correctness around 0.11), a pattern Sections 4.6 and 6.3 attribute primarily to CUAD's task construction rather than retriever quality. On Azure every configuration reaches Recall@5, nDCG@5 and MRR near 1.0 with faithfulness 0.93–0.96: a ceiling effect from a small corpus of short, well-separated documents. These three regimes (a discriminating benchmark, a near-intractable one and a saturated one) frame the statistics that follow. Table 4.1b juxtaposes the corpora on the adaptive configuration's two most informative metrics.

Corpus	Recall@5 (adaptive)	Faithfulness (adaptive)	Regime
FinanceBench	0.633	0.814	Discriminating
CUAD	0.087	0.618	Floor (task-construction limitation)
Azure	1.000	0.950	Ceiling (saturated)

Table 4.1b. The three evaluation regimes, illustrated with the adaptive-hybrid configuration.

Only FinanceBench occupies a middle band where methods separate; CUAD's floor and Azure's ceiling compress the configurations, so an absence of significant differences there is uninformative, not evidence of equivalence. The narrative below draws its positive findings from FinanceBench and treats CUAD and

Azure as generalisation tests.

4.2 RQ1: Does Hybrid Retrieval Beat Single-Method Retrieval?

RQ1 compares the RRF static hybrid against each single-method baseline on the retrieval metrics, per question, with Holm–Bonferroni correction within each dataset. On FinanceBench the static hybrid's improvement over sparse BM25 is large and clear: Recall@5 rises from 0.300 to 0.593, MRR from 0.210 to 0.389 and nDCG@5 from 0.247 to 0.463, each significant after correction (Holm-adjusted $p < 0.001$) with Cohen's dz between 0.53 and 0.62, a medium effect. Table 4.2 reports this comparison on the three non-redundant retrieval metrics.

Metric	BM25 (a)	Hybrid (b)	Diff	95% CI	d_z	p (Holm)
Recall@5	0.300	0.593	+0.293	[+0.220, +0.367]	0.62	<0.001
MRR	0.210	0.389	+0.179	[+0.126, +0.235]	0.53	<0.001
nDCG@5	0.247	0.463	+0.216	[+0.160, +0.274]	0.61	<0.001

Table 4.2. RQ1, FinanceBench: static hybrid (RRF) versus sparse BM25 (paired, n = 150, Holm-corrected). 95% CI is the bootstrap interval (10,000 resamples) for the mean difference.

Precision@5 and MAP@5 are omitted as exact rescalings of Recall@5 and MRR (Section 3.6; Appendix A).

The comparison against the dense baseline tells a different story: on FinanceBench the static hybrid was numerically worse than dense retrieval (Recall@5 0.59 versus 0.68, 95% CI for the difference [−0.160, −0.013]; nDCG@5 difference −0.107), but no difference survived Holm correction (adjusted p = 0.067–0.40). RQ1 is therefore supported only conditionally: hybrid retrieval reliably rescues weak sparse retrieval with large, significant gains over BM25, but does not beat a strong dense retriever here. On CUAD and Azure the floor and ceiling leave no room to distinguish methods (Figure 4.1). RQ1 is thus supported in the specific, defensible form "hybrid significantly outperforms sparse", not the stronger "hybrid significantly outperforms both baselines" implied by the original hypothesis.

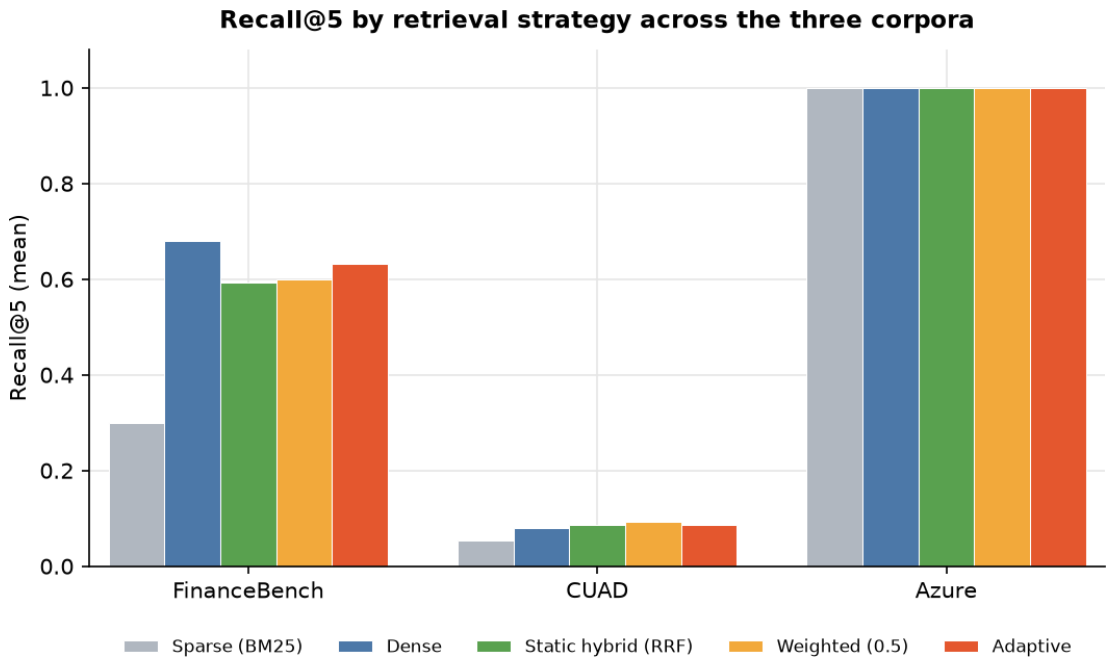


Figure 4.1. Mean Recall@5 by retrieval strategy across the three corpora. FinanceBench discriminates between methods (sparse lowest, dense highest); CUAD sits near the floor — primarily a task-construction limitation (Section 4.6) — and Azure at the ceiling. Cited variants are omitted as their retrieval is identical to the non-cited configurations. Source: author's own analysis.

4.3 RQ2: Does Citation Grounding Improve Faithfulness?

RQ2 compares the cited and non-cited versions of the hybrid configurations on faithfulness and answer correctness. The result is null, with a slight negative tendency: no citation comparison achieves significance on faithfulness or correctness on any dataset after Holm–Bonferroni correction, and the effect is frequently negative, with citations tending to lower faithfulness slightly. The clearest case is CUAD, where faithfulness falls from 0.703 to 0.634 when citations are added to the static hybrid (Cohen's dz = −0.20; raw Wilcoxon p = 0.002), a small effect that does not survive multiple-comparison correction (adjusted p = 0.060). On FinanceBench and Azure the citation effect on faithfulness is small and non-significant in both directions (Table 4.3).

Dataset	Comparison	Faith. (no cite)	Faith. (cite)	Diff	95% CI	p (Holm)
FinanceBench	static_hybrid	0.793	0.767	−0.025	[−0.069, +0.019]	1.000
FinanceBench	adaptive_hybrid	0.814	0.806	−0.008	[−0.057, +0.040]	1.000
CUAD	static_hybrid	0.703	0.634	−0.068	[−0.122, −0.014]	0.060
CUAD	adaptive_hybrid	0.618	0.603	−0.015	[−0.065, +0.035]	1.000
Azure	static_hybrid	0.930	0.943	+0.014	[−0.026, +0.057]	1.000

Dataset	Comparison	Faith. (no cite)	Faith. (cite)	Diff	95% CI	p (Holm)
Azure	adaptive_hybrid	0.950	0.936	−0.013	[−0.074, +0.042]	1.000

Table 4.3. RQ2: effect of citation grounding on faithfulness (paired, Holm-corrected within dataset). 95% CI is the bootstrap interval for the mean difference (cite − no-cite). CUAD adaptive pair: n = 149.

Verifiability, by contrast, is achieved. The invalid-citation rate is 0.000 for every cited configuration on every dataset: GPT-4.1 never emitted a marker pointing outside the supplied passage set. Citation coverage, computed over non-refusal answers, ranged from 0.43 on CUAD and 0.72–0.74 on FinanceBench to 0.81 on Azure: the model cites reliably within range but does not annotate every sentence. Citations left answer relevancy within ±0.01 on FinanceBench and Azure and lowered it by 0.02–0.04 on CUAD. One pair runs counter to the general direction: on FinanceBench the fixed-weight hybrid shows faithfulness rising when citations are added (0.756 to 0.800; raw Wilcoxon p = 0.015). This pair falls outside the RQ2 family defined in Section 3.7, so the p-value is uncorrected; it is reported for transparency. The divergent sign reinforces the central RQ2 conclusion: the citation effect is not stable across fusion methods or corpora.

An exploratory check restricted to questions answered in both configurations leaves FinanceBench unchanged (−0.024 and +0.004) but shows lower faithfulness with citations on CUAD in both pairs (−0.090, n = 108, raw p < 0.001; −0.080, n = 106, raw p = 0.001).

The overall interpretation is that citation grounding, under this prompt and generator, is a useful auditability mechanism, making claims traceable to in-range passages with no out-of-range markers (structural validity, not verified semantic support), but not a faithfulness intervention: on this evidence it does not make the underlying answer reliably more grounded, and its small effect varies in direction across corpora and fusion methods (Figure 4.2). RQ2 is therefore not supported for faithfulness, though the verifiability sub-claim is.

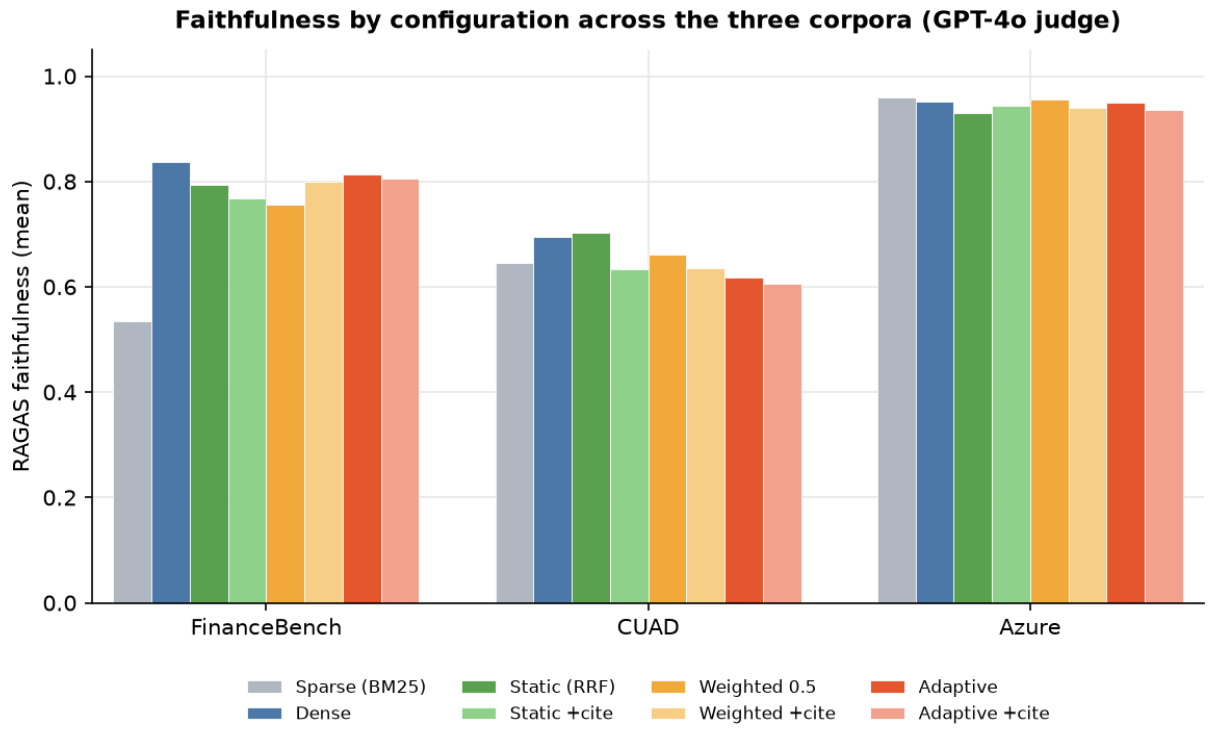


Figure 4.2. Mean RAGAS faithfulness (GPT-4o judge) by configuration across the three corpora. The cite and non-cite variant of each hybrid are shown adjacently; the citation effect is small and inconsistent in sign across corpora and fusion methods (RQ2). Source: author's own analysis.

4.4 RQ3: Does Adaptive Weighting Beat a Fixed Weight?

RQ3 is the study's central question, and the comparison is controlled by design (Section 3.1). On FinanceBench, adaptive weighting produces a statistically significant improvement in retrieval ranking: MRR rises from 0.413 to 0.480 and nDCG@5 from 0.493 to 0.560, with Cohen's dz of 0.38–0.39 and Holm-adjusted p ≈ 0.0004. The same pattern holds for the cited pair, since routing is identical. Recall@5 does not change significantly (0.600 versus 0.633), suggesting that adaptive weighting mainly reorders the retrieved set, placing relevant documents higher rather than retrieving more of them (Table 4.4).

With each gold document counted once, nDCG@5 falls slightly (fixed-weight 0.460, adaptive 0.519) but no verdict changes: the adaptive gain remains significant (+0.058, Holm-adjusted p = 0.0008).

Metric	Fixed 0.5 (a)	Adaptive (b)	Diff	95% CI	d_z	p (Holm)
MRR (= MAP@5)	0.413	0.480	+0.067	[+0.040, +0.095]	0.39	0.0004
nDCG@5	0.493	0.560	+0.068	[+0.040, +0.097]	0.38	0.0004
Recall@5	0.600	0.633	+0.033	[+0.000, +0.067]	0.16	0.82 (n.s.)
Faithfulness	0.756	0.814	+0.059	[+0.013, +0.107]	0.20	0.24 (n.s.)

Table 4.4. RQ3, FinanceBench: adaptive weighting versus fixed-weight (0.5) hybrid (paired, n = 150, Holm-corrected). 95% CI is the bootstrap interval for the mean difference (adaptive − fixed). MAP@5 is omitted as identical to MRR under single-gold relevance (Section 3.6).

However, the gain does not propagate to answer quality. None of the RAGAS dimensions (faithfulness, answer relevancy, answer correctness or context recall) differs significantly between the fixed and adaptive hybrids on any dataset after correction. The faithfulness improvement on FinanceBench (0.756 to 0.814) is in the expected direction and non-trivial, but does not reach significance (adjusted p = 0.24). On CUAD and Azure, adaptive weighting yields no significant improvement on any metric. RQ3 is therefore supported in a specific form (adaptive weighting significantly improves retrieval ranking on FinanceBench) but not in the stronger form the hypothesis proposed, that it would improve faithfulness and correctness. The improvement is real but confined to retrieval ordering, and to the one corpus where the router's features vary.

4.5 Router Behaviour

The routing logs explain why RQ3's effect is corpus-specific: the distribution of per-query dense weights differs sharply across corpora. On FinanceBench the router exercises a wide range of weights (0.40 to 0.71, mean 0.56, SD 0.07), labelling a majority of queries semantic, and this is the corpus where adaptive weighting produces significant ranking gains. On CUAD it labels 148 of 150 queries lexical and pins the dense weight near 0.43 (SD 0.01), so it behaves almost like a fixed-weight hybrid and produces no significant difference from the baseline. On Azure the weight hovers near 0.49 with moderate spread (SD 0.06), and route labels are mixed (Figure 4.3). The router therefore helps only where query characteristics vary, and the significant RQ3 result is driven by the single corpus on which the routing signal is informative,

examined in Section 5.4.

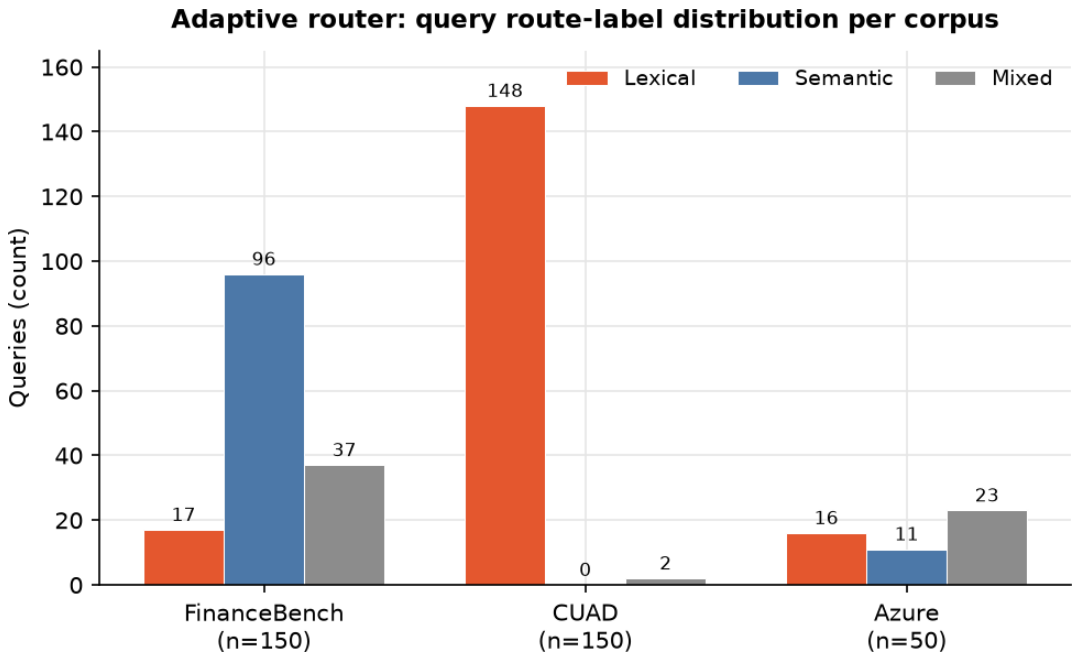


Figure 4.3. Distribution of adaptive-router route labels per corpus. The router varies its decision on FinanceBench (a majority of queries labelled semantic) but labels 148 of 150 CUAD queries lexical, collapsing toward the fixed-weight baseline it is meant to beat (Section 5.4). Source: author's own analysis.

4.6 Error Analysis

Low-faithfulness answers (below 0.70) were categorised into four error types, of which only three were populated: citation mismatch, operationalised as an out-of-range marker, stayed empty because the invalid-citation rate was zero on every dataset (Section 4.3). The dominant category differs revealingly by corpus. On CUAD the overwhelming failure mode is retrieval failure (41 to 69 cases per non-adaptive configuration with no relevant document in the top five), and the "routing-associated retrieval failure" category for adaptive configurations (61 to 75 cases) is the same failure relabelled. The audit logs confirm why (Section 3.5): CUAD's clause-extraction prompts name no contract, so the retriever cannot in principle select the correct document; the low CUAD retrieval metrics are strongly attributable to task construction rather than retriever weakness.

On FinanceBench the picture is healthier: retrieval failures are far fewer for hybrid and dense configurations (14 to 22 cases) than for sparse BM25 (61 cases), and most are refusals the judge scored as unfaithful (57 of BM25's 61); the remaining low-faithfulness cases (15 to 26 per dense or hybrid configuration) are unsupported synthesis: relevant evidence was retrieved but the generator produced a claim the context did not fully support. This is consistent with retrieval and generation failing independently (Section 5.2). Refusal rates track retrieval difficulty, reaching 74% for BM25 on FinanceBench against 37–47% for the dense and hybrid configurations, and 0% throughout on the saturated Azure corpus.

Three manually inspected cases make these categories concrete and offer a limited human check. First, asked for 3M's FY2018 capital expenditure, the cited static hybrid answered "$1,577 million. [4]" (faithfulness 1.00, coverage 1.00). Second, asked whether 3M was capital-intensive on FY2022 data, the dense and static-hybrid configurations retrieved the correct filing and reported net property, plant and equipment of $9,178m against total assets of $46,455m, but added the label "capital-intensive", which the passage does not state and the reference answer disputes; the judge scored both 0.40 yet gave similar answers from five other configurations 1.00. Third, a CUAD prompt naming no contract retrieved unrelated agreements and the model correctly refused, yet across CUAD the judge scored 188 of 288 such refusals 0.0. The judge therefore flags over-reach inconsistently and penalises some correct abstentions, so human annotation remains the priority check (Section 5.5).

5. DISCUSSION

This chapter interprets the results against the guiding hypothesis and the literature, and sets out their limits.

5.1 Interpreting the Findings Against the Hypothesis

The guiding hypothesis (that adaptive hybrid retrieval with citation grounding would achieve higher faithfulness than single-method baselines, and that citation would improve verifiability without reducing relevance) is supported only in part, and the pattern of partial support is itself the most substantive finding. Hybrid retrieval delivers large, significant gains over sparse retrieval but does not surpass a strong dense retriever, making it a remedy for sparse-only retrieval rather than a uniform improvement. Adaptive weighting significantly improves retrieval ranking but not answer quality, and only where query characteristics vary enough for the router to act. Citation grounding delivers clean verifiability but no consistent faithfulness benefit, its small effect varying in sign across corpora and fusion methods.

This partly null pattern is nevertheless informative: it delineates where each component helps and provides evidence relevant to organisational decisions about engineering investment. The strongest inferential claims therefore rest on FinanceBench; CUAD and Azure serve as robustness and stress tests, not equally discriminating comparisons.

5.2 Why Retrieval Gains Do Not Become Faithfulness Gains

Why does a significant improvement in retrieval ranking (RQ3) not improve faithfulness? The error analysis suggests a mechanism: in the dense and hybrid configurations unsupported synthesis was about as common as retrieval failure (Section 4.6), and most retrieval failures were refusals. Adaptive weighting acts on passage ordering; it cannot stop a generator over-reaching once adequate evidence is present. The fixed and adaptive hybrids share on average 4.4 of their five passages, so adaptivity mostly reorders rather than replaces the evidence the generator sees. Under the conditions studied here, improving faithfulness further appears to depend at least as much on generation, motivating interventions such as better prompting, claim-level verification and calibrated abstention.

5.3 The Meaning of the Citation Result

The RQ2 result (verifiability without faithfulness improvement) has a clear practical reading. Inline citations reliably produced traceable answers with zero invalid markers, a property particularly valuable in regulated settings. But citations did not make claims more grounded: on CUAD the per-sentence requirement coincided with a modest faithfulness fall, concentrated among answered questions, plausibly because it pressures the model to attach a marker even when no passage cleanly supports the sentence. That this is corpus-specific is shown by the FinanceBench fixed-weight pair, where citations coincided with a rise; the instruction perturbs faithfulness slightly and unpredictably rather than degrading it systematically. Citation should therefore be treated as an auditability layer decoupled from faithfulness, with coverage and invalid-rate monitored alongside it.

5.4 The Router's Corpus Dependence

The significant RQ3 effect appearing only on FinanceBench is consistent with how much the router's input signal varies. On FinanceBench, queries range from precise numeric look-ups to interpretive questions, so the features vary and the router moves the weight meaningfully; on CUAD the near-identical clause-extraction template makes queries look alike, so it settles on a near-constant weight and collapses toward

the baseline. It behaves as designed (adapting when there is variation, defaulting sensibly when there is not), but its benefit is contingent on query heterogeneity. For an enterprise with a mixed query stream the FinanceBench result is relevant; for templated queries adaptivity offers little over a well-chosen fixed weight , a scoping condition rather than a defect. The logs also show that on FinanceBench the weight tracked query length closely (Spearman $\rho = 0.66$), so the ranking gain may owe as much to length as to lexical–semantic intent; narrower bounds were not tested.

5.5 Methodological Considerations

Two design choices shaped what the study could detect. Pairing was clearly beneficial: because every configuration answered the same questions, question difficulty is differenced out, making the tests more powerful than an unpaired design and helping the RQ1 and RQ3 ranking effects emerge on 150 questions. Holm–Bonferroni correction, by contrast, is conservative: it left the CUAD citation effect a near-miss (adjusted $p = 0.060$) and the FinanceBench faithfulness gain non-significant ($p = 0.24$); this is the correct trade-off for a confirmatory study, but some non-significant differences may reflect limited power rather than true nulls.

The second choice is reliance on automated, LLM-judged metrics, mitigated by a separate judge and an error analysis grounded in the audit logs. The scores inherit the judge's biases: because judge (GPT-4o) and generator (GPT-4.1) share a provider and family, a distinct judge reduces but does not eliminate self-preference bias (Panickssery, Bowman and Feng, 2024); the answer-quality conclusions concern a well-specified proxy, not ground truth. The manual cases in Section 4.6 show this proxy is not always consistent, so human annotation of a stratified subsample is the most important next check.

5.6 Threats to Validity

Four classes of validity threat bound the conclusions. Construct validity: faithfulness is operationalised through an LLM judge rather than adjudicated hallucination ground truth, so the study measures a well-specified proxy, with the stratified error analysis (Section 4.6) as the check on that proxy. Internal validity: the adaptive gain could in principle reflect the weighted-fusion or score-normalisation step rather than routing itself, a threat the identically-fused fixed-weight baseline substantially mitigates, leaving query-adaptivity as the only deliberately varied factor (Section 3.1). External validity: FinanceBench is evaluated over evidence passages rather than full filings, and a single generator and dense encoder are used, so generalisation to full-document, multi-model enterprise deployments is not established. Measurement validity: single-gold labelling makes two retrieval metrics redundant (Section 3.6), and the nDCG@5 implementation can exceed 1, although a de-duplicated check changes no verdict (Section 4.4).

5.7 Comparison with the Literature

The retrieval findings are consistent with the broader literature. The strong dense performance and large sparse-to-hybrid gain echo BEIR (Thakur et al., 2021): neither paradigm dominates universally. That the choice of fusion function shapes retrieval quality aligns with Bruch, Gai and Ingber (2024). The adaptive router's corpus-dependent performance extends the Adaptive-RAG findings of Jeong et al. (2024). Both studies suggest that adaptivity helps where queries are heterogeneous; their learned router is optimised from data at the cost of opacity, whereas the heuristic router used here is fully auditable but untuned.

The null citation-faithfulness result contrasts with the training-based gains reported by Nakano et al. (2021) in WebGPT, where accuracy improved because the model was optimised through human feedback

to produce faithful cited answers. Prompting a fixed generator to cite (with no training signal toward faithful citation) changes presentation more than substance. This aligns with Wallat et al. (2025): every marker points to a real passage, yet faithfulness is unchanged. Tamber et al. (2025) provide independent evidence that faithfulness remains fragile even when the relevant context is supplied, consistent with a bottleneck in generation rather than retrieval or citation mechanics.

The study's central empirical finding (that a significant improvement in retrieval ranking does not automatically produce a faithfulness improvement) is consistent with independent work questioning whether retrieval quality drives generation quality. Cuconasu et al. (2024) show that, in a constrained setting, injecting irrelevant passages can improve answer accuracy, undermining the assumption that retrieval purity drives answer quality. The present study's own error analysis points the same way (Section 4.6). A distinctive methodological feature is that, by giving the adaptive and fixed-weight hybrids the same fusion function and comparing them under a paired design, the study isolates the retrieval variable well enough for the dissociation to be observed, which evaluations of retrieval alone (e.g. Thakur et al., 2021) or generation alone cannot show.

6. CONCLUSION

6.1 Summary of Contributions

RQ	Finding	Verdict
RQ1	Hybrid significantly outperforms sparse ($p < .001$, $d_z = 0.53–0.62$); Hybrid significantly outperforms sparse ($p < .001$, $dz = 0.53–0.62$); numerically below dense, not significant after correction	Partially supported
RQ2	No significant faithfulness difference; verifiability clean (no out-of-range markers; coverage 0.43–0.81)	Not supported (faithfulness); supported (verifiability)
RQ3	Ranking improved on FinanceBench ($p < .001$, $dz = 0.38–0.39$); answer quality not significantly different	Partially supported (FinanceBench ranking only)

Table 6.1. Summary of research-question verdicts.

This dissertation asked whether an adaptive hybrid RAG framework with citation grounding makes enterprise question answering measurably more reliable than single-method and static-hybrid baselines. Through a controlled evaluation of eight configurations over three corpora, analysed with a paired, Holm-corrected protocol, it reached a nuanced answer. Hybrid retrieval substantially outperforms sparse retrieval (RQ1) but does not beat a strong dense retriever. Citation grounding produces structurally valid, in-range citations that improve traceability without verifying semantic support, and does not significantly change faithfulness, which was slightly lower in some comparisons (RQ2). Query-adaptive weighting significantly improves retrieval ranking over an identically-fused fixed-weight hybrid on the corpus with heterogeneous queries (RQ3), though the gain does not extend to answer quality or to corpora with homogeneous queries. Table 6.1 summarises the verdicts.

The methodological contributions set out in Section 1.4 stand independently of the effect sizes, and together indicate that ranking improvements should not be assumed to yield faithfulness improvements, and that citation is best treated as auditability rather than a hallucination remedy.

6.2 Implications for Practice

For practitioners, three recommendations follow. First, if a system relies on sparse retrieval, adding a dense component is the highest-value change; a strong dense retriever alone may suffice on semantically phrased corpora, and prior work suggests hybrid retrieval is best justified where lexical precision (exact figures, clause references) cannot be sacrificed (Thakur et al., 2021). Second, a transparent adaptive router is a low-risk addition, but its benefit should be validated against the actual query distribution, since it offers little on homogeneous workloads. Third, inline citation should be deployed for auditability, not relied upon to reduce hallucination; faithfulness should be pursued at generation.

6.3 Limitations

Several limitations bound the findings. The CUAD evaluation is compromised by task construction: the clause-extraction prompts do not identify which contract is intended, making single-document retrieval

effectively impossible. The Azure corpus is saturated (near-perfect retrieval for all methods), providing only a robustness check rather than a discriminating comparison. FinanceBench uses evidence passages instead of full filings, approximating an open-book setting and understating full-document retrieval difficulty. The router's benefit was demonstrated on a single corpus, so its generality is not established. Faithfulness relies on a single LLM judge that scored refusals and near-identical answers inconsistently (Sections 3.6 and 4.6). A single generator and dense model (all-MiniLM-L6-v2) were used; that model truncates input beyond 256 word pieces (Sentence Transformers, no date), so many longer chunks were embedded only in part, and chunk size and overlap were not ablated. Temperature was fixed at zero throughout, so whether routing and citation effects behave differently under sampling remains untested.

6.4 Future Work

Four directions follow. First, because faithfulness failures on FinanceBench were dominated by unsupported synthesis rather than retrieval failure, the highest-value extension is generation-stage intervention (claim-level verification against retrieved evidence, or calibrated abstention) under the same confound-controlled design. Second, the router should be tested across corpora with varied query heterogeneity, and its weights lightly tuned or replaced with a small learned router while preserving auditability. Third, the citation mechanism should be re-examined with training or preference optimisation toward faithful citation, testing whether the WebGPT-style benefit can be recovered without sacrificing transparency. Fourth, the evaluation should span multiple generators and dense encoders to map the retrieval–generation interaction. Each builds on the reproducible framework delivered here.

6.5 Concluding Remarks

The study set out to test whether adaptive fusion and citation grounding make enterprise QA more reliable. Its answer is measured: adaptivity improves how evidence is ranked and citation improves how answers can be audited, but neither measurably reduced hallucination, and the remaining faithfulness failures arose at least as often at generation as at retrieval. Demonstrating this is the dissertation's contribution, and it points the next effort at the generation stage.

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APPENDIX A: CONFIGURATION AND HYPERPARAMETERS

The following settings were held constant across all eight configurations unless the configuration definition varied them. Chunk size: 512 tokens; chunk overlap: 64 tokens (tiktoken cl100k_base). Candidate pool per retriever (retrieve_k): 20; passages returned to the generator (top_k): 5. Dense embedding model: all-MiniLM-L6-v2 (sentence-transformers), L2-normalised, indexed with FAISS inner-product search. RRF constant k: 60. Adaptive dense-weight bounds: [0.2, 0.8]; fixed-weight hybrid dense weight: 0.5. Generator: GPT-4.1, temperature 0. RAGAS judge: GPT-4o, temperature 0; judge embedding model: text-embedding-3-large. Bootstrap resamples: 10,000; random seed: 42 (bootstrap and CUAD sampling).

Redundant retrieval metrics. The pipeline additionally computes Precision@5 and MAP@5, which are omitted from the main-text tables (Section 4.2) because they are exact algebraic rescalings of Recall@5 and MRR respectively under the single-gold-document labelling of these corpora, and so add no independent evidence. For completeness, the FinanceBench values are: Precision@5 = Recall@5 / 5 for every configuration (for example sparse_bm25 = 0.060, dense = 0.136, static_hybrid = 0.119, adaptive_hybrid = 0.127); and MAP@5 = MRR for every configuration (for example static_hybrid_weighted = 0.413, adaptive_hybrid = 0.480). The identities hold exactly on FinanceBench and CUAD and break only on the three two-gold-document Azure questions, where Precision@5 reaches 0.4 and MAP@5 diverges marginally from MRR.

Decoding temperature. Temperature zero was chosen for the reasons given in Section 3.4. For comparison, the studies cited in this dissertation report the following decoding settings. Lewis et al. (2020) use greedy decoding for question answering and beam search of width four for generation tasks. Asai et al. (2024) combine a segment-level beam of width two with token-level greedy decoding. Cuconasu et al. (2024) use greedy generation. Islam et al. (2023) evaluate FinanceBench at temperature 0.01. Nakano et al. (2021) sample WebGPT at temperature 0.8, tuned by human evaluation, but do so to produce up to sixty-four candidate answers for a learned reward model to re-rank, making sampling diversity an input to a later selection stage rather than the final output; no such re-ranker operates here. Decoding settings could not be located in Jeong et al. (2024), Edge et al. (2024) or Es et al. (2024).

APPENDIX B: DATASETS

FinanceBench: 144 corpus documents, 150 questions (source: PatronusAI/financebench; evidence passages used as corpus, content-hash de-duplicated). CUAD: 102 test-split contracts, 150 seeded-sample questions (source: TheAtticusProject/cuad, test split). Azure: 55 documents and 50 manually constructed questions (custom enterprise-documentation corpus). The 50 questions comprise 22 lexical (exact-term or numeric) and 28 semantic (interpretive) items; each carries a reference answer and is mapped to its gold source document (47 to a single document, three to two), with every mapping checked against the source text. A small sample corpus (6 documents, 6 questions) ships with the implementation for a free, no-cost reproducibility check.

APPENDIX C: DIGITAL RESOURCES AND CODE REPOSITORY

The complete implementation is a single Python module (ahrag.py) with pinned dependency versions (requirements.txt). It supports a free dry-run mode (data and chunking validation, no API keys), a limited live smoke test, one-time dataset builders for the public benchmarks, the full eight-configuration study and an analysis-only mode that regenerates all summary tables, statistics, error analysis and per-metric charts from existing result files. Figures 4.1–4.3 were prepared separately from summary.csv and

all_results.csv, and supplementary_checks.py reproduces the additional checks reported in Sections 3.6, 4.3, 4.4, 4.6, 5.2, 5.4 and 6.3. Per-question metrics, combined results, NaN-aware summary means, the full Wilcoxon/t-test/effect-size/bootstrap/Holm statistics table, error-category counts and per-metric bar charts are written to a structured results directory, alongside per-question audit logs recording every retrieval decision, prompt, answer, parsed citation and routing decision.

The full source code, pinned dependencies (requirements.txt), dataset builders and the complete results directory accompany this dissertation as the submitted primary-data archive (F517783_PRIMARY_DATA.zip). The archive is self-contained: it carries the analysis programme, the four compiled corpora, every results file and the complete per-question audit logs, so every table, statistic and figure can be re-derived from the stored outputs without an external repository; regenerating the answers themselves requires an OpenAI API key to reproduce the results.